What is the right geometry for minimization of MMD?

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August 14, 2026

1 Introduction

Given a functional $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$, its gradient-flow dynamics depend on the choice of geometry on the underlying space of measures. One of the most commonly used geometries is the Wasserstein-2 geometry, induced by the Wasserstein-2 distance. Its associated Onsager operator, which is the inverse of the Riemannian metric tensor, is given by $\mathbb{K}_W(\rho) : T^*_\rho\mathcal{M}^+ \to T_\rho\mathcal{M}^+, \xi \mapsto -\operatorname{div}(\rho\nabla\xi)$. Hence, the Wasserstein gradient flow of $\mathcal{F}$ is

$$\partial_t\mu_t = \nabla \cdot \left(\mu_t\nabla\mathcal{F}'[\mu_t]\right). \tag{W-GF}$$

Similarly, the Fisher–Rao geometry is induced by the Hellinger distance, with associated Onsager operator $\mathbb{K}_{\mathrm{FR}}(\rho) : T^*_\rho\mathcal{M}^+ \to T_\rho\mathcal{M}^+, \xi \mapsto \rho, \xi$. Consequently, the Fisher–Rao gradient flow of $\mathcal{F}$ is

$$\partial_t\mu_t = -\mu_t\left(\mathcal{F}'[\mu_t] - \int \mathcal{F}'[\mu_t]\mathrm{d}\mu_t\right). \tag{FR-GF}$$

An interpolation between these two dynamics is given by the well-known Wasserstein–Fisher–Rao gradient flow:

$$\partial_t\mu_t = \alpha\nabla \cdot \left(\mu_t\nabla\mathcal{F}'[\mu_t]\right) - \beta\mu_t\left(\mathcal{F}'[\mu_t] - \int \mathcal{F}'[\mu_t]\mathrm{d}\mu_t\right). \tag{WFR-GF}$$

Recently, a different geometry is considered in Zhu [2024], Zhu and Mielke [2024], Gladin et al. [2024], which uses the MMD as the geometry. The associated Onsager operator is $\mathbb{K}_{\mathrm{MMD}}(\rho) = \mathbb{K}_{\mathrm{FR}}(\rho)\circ\mathcal{K}_\rho^{-1}$. The Interaction-Force Transport (IFT) gradient flow is described as the following:

$$\partial_t\mu_t = -\mathbb{K}_{\mathrm{MMD}}(\mu_t)\mathcal{F}'[\mu_t] = -\mu_t\mathcal{K}_{\mu_t}^{-1}\mathcal{F}'[\mu_t] = -\mathcal{K}^{-1}\mathcal{F}'[\mu_t]. \tag{IFT-GF}$$

Similarly as WFR-GF, an interpolation of WGF and IFT-GF can also be constructed

$$\partial_t\mu_t = \alpha\nabla \cdot \left(\mu_t\nabla\mathcal{F}'[\mu_t]\right) - \beta\mathcal{K}^{-1}\mathcal{F}'[\mu_t]. \tag{WIFT-GF}$$

2 $\mathcal{F} = \mathrm{MMD}$ and geometry$=\mathrm{MMD}$

We focus on MMD due to its interpretation as the worse-case integration error in the unit ball of a RKHS. If $\mathcal{F}$ is the MMD distance $\mathcal{F}(\cdot) = \mathrm{MMD}^2(\cdot, \pi)$, then $\mathcal{F}'[\mu](\cdot) = \int k(x, \cdot)\mathrm{d}(\mu - \pi) = \mathcal{K}(\mu - \pi)$. MSIP uses the WFG-GF, but here we focus on the WIFT-GF. The main reason is that, Gladin et al. [2024] has proved that $\mathrm{MMD}^2\left(\mu_t, \pi\right) \le e^{-2\beta t} \cdot \mathrm{MMD}^2\left(\mu_0, \pi\right)$.

Following Gladin et al. [2024], a way to simulate the WIFT-GF. is via an weighted interacting particle system. Let $\mu_N^{(t)} = \sum_{i=1}^N w_i^{(t)} \delta_{x_i^{(t)}}$ be a weighted empirical measure at iteration t, where the weights do not necessarily sum up to 1. Write $\boldsymbol{X}_t = (x_1^{(t)}, \ldots, x_N^{(t)})$ and $\boldsymbol{w}^{(t)} = (w_1^{(t)}, \ldots, w_N^{(t)})$. Denote $m_\pi = \int k(x, \cdot) \mathrm{d}\pi$. First evolve the Wasserstein gradient term,

$$x_i^{(t+1)} = x_i^{(t)} - \tau\alpha \left[\sum_{j=1}^N w_j^{(t)} \nabla_2 k\left(x_j^{(t)}, x_i^{(t)}\right) - \nabla m_\pi\left(x_i^{(t)}\right) \right]. \tag{1}$$

Then, evolve the IFT gradient term, by solving the following optimal weights: let $\eta = \beta\tau\alpha^{-1}$,

$$\boldsymbol{w}^{(t+1)} = \arg\min_{\boldsymbol{w} \in \mathbb{R}^N} \left\{ \frac{1}{2} \mathrm{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i^{(t+1)}}, \pi \right) + \frac{1}{2\eta} \mathrm{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i^{(t+1)}}, \mu_N^{(t)} \right) \right\}. \tag{2}$$

Interestingly, letting $K^{(t+1)} = K(\boldsymbol{X}_{t+1}) \in \mathbb{R}^{N \times N}$, the weight problem is

$$\min_{\boldsymbol{w}} \left\{ \frac{1}{2} \boldsymbol{w}^\top K^{(t+1)} \boldsymbol{w} - m_\pi(\boldsymbol{X}_{t+1})^\top \boldsymbol{w} + \frac{1}{2\eta} \left(\boldsymbol{w} - \boldsymbol{w}^{(t)}\right)^\top K^{(t+1)} \left(\boldsymbol{w} - \boldsymbol{w}^{(t)}\right) \right\},$$

By definition of kernel quadrature, there is $m_\pi(\boldsymbol{X}_{t+1}) = K^{(t+1)} \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})$. And the solution is

$$\boldsymbol{w}^{(t+1)} = \frac{1}{1+\eta} \boldsymbol{w}^{(t)} + \frac{\eta}{1+\eta} \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1}).$$

Theorem 2.1 (Monotone improvement)**.** *At iteration $t \in \mathbb{N}$, let $\mu_N^{(t)} = \sum_{i=1}^N w_i^{(t)} \delta_{x_i^{(t)}}$ with $\boldsymbol{X}_t = (x_1^{(t)}, \ldots, x_N^{(t)})$, and $\boldsymbol{w}^{(t)} = (w_1^{(t)}, \ldots, w_N^{(t)})$. Let $\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}) = K(\boldsymbol{X})^{-1} m_\pi(\boldsymbol{X})$ denote the kernel quadrature weights associated with the locations $\boldsymbol{X}$.*

Assume that k is twice continuously differentiable, that $K(\boldsymbol{X}_t)$ is positive definite for every t, and that the weights remain uniformly positive, namely $w_i^{(t)} \geq \underline{w} > 0$ for every i, t. Assume moreover that, for every admissible $\boldsymbol{w}$, the map $\boldsymbol{X} \mapsto \mathcal{E}(\boldsymbol{X}, \boldsymbol{w})$ has an L_X-Lipschitz gradient on a set containing all iterates. If $\tau\alpha \leq 2\underline{w}/L_X$, then for every $t \geq 0$,

$$\mathrm{MMD}\left(\mu_N^{(t+1)}, \pi\right) \leq \mathrm{MMD}\left(\mu_N^{(t)}, \pi\right). \tag{3}$$

Proof. For fixed $\boldsymbol{X}$ and $\boldsymbol{w}$, let $g_i(\boldsymbol{X}, \boldsymbol{w}) = \sum_{j=1}^N w_j \nabla_2 k(x_j, x_i) - \nabla m_\pi(x_i)$. Define the MMD energy by

$$\mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2} \mathrm{MMD}^2 \left(\sum_{i=1}^N w_i \delta_{x_i}, \pi \right), \tag{4}$$

Differentiating the MMD energy with respect to x_i and using symmetry of the kernel gives $\nabla_{x_i} \mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = w_i g_i(\boldsymbol{X}, \boldsymbol{w})$. Hence, the position update can be written as $x_i^{(t+1)} = x_i^{(t)} - \frac{\tau\alpha}{w_i^{(t)}} \nabla_{x_i} \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}^{(t)})$. Thus, the Wasserstein substep is a diagonally preconditioned gradient step for $\boldsymbol{X} \mapsto \mathcal{E}(\boldsymbol{X}, \boldsymbol{w}^{(t)})$. By the L_X-smoothness of $\mathcal{E}(\cdot, \boldsymbol{w}^{(t)})$, the descent lemma gives

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t)}) \leq \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}^{(t)}) + \sum_{i=1}^N \left\langle \nabla_{x_i} \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}^{(t)}), x_i^{(t+1)} - x_i^{(t)} \right\rangle + \frac{L_X}{2} \sum_{i=1}^N \left\| x_i^{(t+1)} - x_i^{(t)} \right\|^2$$

$$= \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}^{(t)}) - \tau\alpha \sum_{i=1}^{N} \left(w_i^{(t)} - \frac{L_X \tau \alpha}{2} \right) \left\| g_i(\boldsymbol{X}_t, \boldsymbol{w}^{(t)}) \right\|^2. \tag{5}$$

Since $w_i^{(t)} \geq \underline{w}$ and $\tau\alpha \leq 2\underline{w}/L_X$, we have $w_i^{(t)} - L_X\tau\alpha/2 \geq 0$. Therefore, $\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t)}) \leq \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}^{(t)})$. This proves that the particle update in Eq. (1) is decreasing the MMD value.

We next consider the weight update. For fixed $\boldsymbol{X}$, the dependence of the MMD energy on $\boldsymbol{w}$ is quadratic: $\mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2}\boldsymbol{w}^\top K(\boldsymbol{X})\boldsymbol{w} - m_\pi(\boldsymbol{X})^\top \boldsymbol{w} + \text{const}$. Since $K(\boldsymbol{X})$ is positive definite, the unique minimizer is $\boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}) = K(\boldsymbol{X})^{-1} m_\pi(\boldsymbol{X})$. Completing the square gives

$$\mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = \mathcal{E}\left(\boldsymbol{X}, \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X})\right) + \frac{1}{2} \left(\boldsymbol{w} - \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X})\right)^\top K(\boldsymbol{X}) \left(\boldsymbol{w} - \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X})\right). \tag{6}$$

By the weight update,

$$\boldsymbol{w}^{(t+1)} - \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_{t+1}) = \frac{1}{1+\eta} \left(\boldsymbol{w}^{(t)} - \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_{t+1}) \right). \tag{7}$$

On the other hand, for fixed particle locations $\boldsymbol{X}_{t+1}$, the KQ decomposition gives

$$\begin{aligned} &\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t+1)}) - \mathcal{E}\left(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_{t+1})\right) \\ &\quad = \frac{1}{2} \left(\boldsymbol{w}^{(t+1)} - \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_{t+1}) \right)^\top K^{(t+1)} \left(\boldsymbol{w}^{(t+1)} - \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_{t+1}) \right). \end{aligned} \tag{8}$$

Substituting (7) into (8) yields

$$\begin{aligned} &\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t+1)}) - \mathcal{E}\left(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_{t+1})\right) \\ &\quad = \frac{1}{2(1+\eta)^2} \left(\boldsymbol{w}^{(t)} - \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_{t+1}) \right)^\top K^{(t+1)} \left(\boldsymbol{w}^{(t)} - \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_{t+1}) \right) \\ &\quad = \frac{1}{(1+\eta)^2} \left[\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t)}) - \mathcal{E}\left(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_{t+1})\right) \right]. \end{aligned} \tag{9}$$

Since $(1+\eta)^{-2} \leq 1$ and $\boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_{t+1})$ minimizes $\boldsymbol{w} \mapsto \mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w})$, it follows that $\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t+1)}) \leq \mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t)})$. Combining the two substeps yields

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t+1)}) \leq \mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t)}) \leq \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}^{(t)}). \tag{10}$$

Hence, $\mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}^{(t)})$ is nonincreasing, and the proof is concluded. □

Lemma 2.2 (Uniform control of the kernel quadrature residual)**.** *Suppose that the assumptions of Theorem 2.1 hold and that the initial weights are chosen as* $\boldsymbol{w}^{(0)} = \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_0)$. *Then, for every* $t \geq 0$, $\mathrm{MMD}\left(\sum_{i=1}^N w_{\text{KQ},i}(\boldsymbol{X}_t)\delta_{x_i^{(t)}}, \pi\right) \leq \mathrm{MMD}\left(\sum_{i=1}^N w_{\text{KQ},i}(\boldsymbol{X}_0)\delta_{x_i^{(0)}}, \pi\right)$.

Proof. Recall the definition that $\mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2}\mathrm{MMD}^2\left(\sum_{i=1}^N w_i \delta_{x_i}, \pi\right)$. For any fixed particle locations $\boldsymbol{X}_t$, the kernel quadrature weights $\boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_t)$ minimize the MMD energy over the weights. Hence, $\mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_t)\right) \leq \mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}^{(t)}\right)$. On the other hand, by the monotonicity of the WIFT iteration established in Theorem 2.1, we have $\mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}^{(t)}\right) \leq \mathcal{E}\left(\boldsymbol{X}_0, \boldsymbol{w}^{(0)}\right)$. Since $\boldsymbol{w}^{(0)} = \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_0)$, we have $\mathcal{E}(\boldsymbol{X}_0, \boldsymbol{w}^{(0)}) = \mathcal{E}(\boldsymbol{X}_0, \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_0))$. Combining the previous inequalities gives

$$\mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_t)\right) \leq \mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}^{(t)}\right) \leq \mathcal{E}\left(\boldsymbol{X}_0, \boldsymbol{w}^{(0)}\right) = \mathcal{E}\left(\boldsymbol{X}_0, \boldsymbol{w}_{\text{KQ}}(\boldsymbol{X}_0)\right). \tag{11}$$

The proof is thus concluded. □

Theorem 2.3 (Convergence to the kernel quadrature error)**.** *Suppose that the assumptions of Theorem 2.1 hold. Then, the WIFT iterates satisfy*

$$\mathrm{MMD}^2(\mu_N^{(t)}, \pi) \leq \frac{1}{(1+\eta)^{2t}} \mathrm{MMD}^2(\mu_N^{(0)}, \pi) + \left(1 - \frac{1}{(1+\eta)^{2t}}\right) \mathrm{MMD}^2 \left(\sum_{i=1}^{N} w_{\mathrm{KQ},i}(\boldsymbol{X}_0)\delta_{x_i^{(0)}}, \pi\right).$$

Proof. From the exact contraction of the weight update in Eq. (9), together with descent of the Wasserstein position update, we have

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t+1)}) \leq \frac{1}{(1+\eta)^2}\mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}^{(t)}) + \left(1 - \frac{1}{(1+\eta)^2}\right) \mathcal{E}\left(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right). \tag{12}$$

By Lemma 2.2, $\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})) \leq \mathcal{E}(\boldsymbol{X}_0, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0))$, and therefore

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}^{(t+1)}) \leq \frac{1}{(1+\eta)^2}\mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}^{(t)}) + \left(1 - \frac{1}{(1+\eta)^2}\right) \mathcal{E}\left(\boldsymbol{X}_0, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)\right). \tag{13}$$

The proof is concluded. □

Remark 2.4. *Theorem 2.3 separates the optimization error from the statistical error of the initial kernel quadrature rule. In particular, if* $\mathrm{MMD}(\sum_{i=1}^N w_{\mathrm{KQ},i}(\boldsymbol{X}_0)\delta_{x_i^{(0)}}, \pi) = O_{\mathbb{P}}(N^{-s/d})$*, then*

$$\begin{aligned}
\mathrm{MMD}^2(\mu_N^{(t)}, \pi) &\leq \frac{1}{(1+\eta)^{2t}} \mathrm{MMD}^2(\mu_N^{(0)}, \pi) + O_{\mathbb{P}}(N^{-2s/d}),\\
\mathrm{MMD}(\mu_N^{(t)}, \pi) &\leq \frac{1}{(1+\eta)^{t}} \mathrm{MMD}(\mu_N^{(0)}, \pi) + O_{\mathbb{P}}(N^{-s/d}).
\end{aligned}$$

Thus, the optimization error decays geometrically until it reaches the kernel quadrature error floor. If, in addition, $\mathrm{MMD}(\mu_N^{(0)}, \pi) = O_{\mathbb{P}}(1)$*, then choosing* $t \geq \frac{s}{d\log(1+\eta)} \log N$ *is sufficient to obtain* $\mathrm{MMD}(\mu_N^{(t)}, \pi) = O_{\mathbb{P}}(N^{-s/d})$*.*

References

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